

Interacting Multipoles of Rank 2 and 4:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta}=R_z^{-13}\cdot\frac{1}{315}\cdot$$

$$\begin{aligned} &+105\cdot R_z^4\cdot R_\beta\cdot R_\epsilon\cdot\delta(\alpha,\gamma)\cdot\delta(\delta,\zeta)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta}\\ &+105\cdot R_z^4\cdot R_\beta\cdot R_\epsilon\cdot\delta(\alpha,\delta)\cdot\delta(\gamma,\zeta)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta}\\ &+105\cdot R_z^4\cdot R_\beta\cdot R_\epsilon\cdot\delta(\alpha,\zeta)\cdot\delta(\gamma,\delta)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta}\\ &+105\cdot R_z^4\cdot R_\beta\cdot R_\zeta\cdot\delta(\alpha,\gamma)\cdot\delta(\delta,\epsilon)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta}\\ &+105\cdot R_z^4\cdot R_\beta\cdot R_\zeta\cdot\delta(\alpha,\delta)\cdot\delta(\gamma,\epsilon)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta}\\ &+105\cdot R_z^4\cdot R_\beta\cdot R_\zeta\cdot\delta(\alpha,\epsilon)\cdot\delta(\gamma,\delta)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta}\\ &+105\cdot R_z^4\cdot R_\gamma\cdot R_\delta\cdot\delta(\alpha,\beta)\cdot\delta(\epsilon,\zeta)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta}\\ &+105\cdot R_z^4\cdot R_\gamma\cdot R_\delta\cdot\delta(\alpha,\epsilon)\cdot\delta(\beta,\zeta)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta}\\ &+105\cdot R_z^4\cdot R_\gamma\cdot R_\delta\cdot\delta(\alpha,\zeta)\cdot\delta(\beta,\epsilon)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta}\\ &+105\cdot R_z^4\cdot R_\gamma\cdot R_\epsilon\cdot\delta(\alpha,\beta)\cdot\delta(\delta,\zeta)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta}\\ &+105\cdot R_z^4\cdot R_\gamma\cdot R_\epsilon\cdot\delta(\alpha,\delta)\cdot\delta(\beta,\zeta)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta}\\ &+105\cdot R_z^4\cdot R_\gamma\cdot R_\epsilon\cdot\delta(\alpha,\zeta)\cdot\delta(\beta,\delta)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta}\\ &+105\cdot R_z^4\cdot R_\gamma\cdot R_\zeta\cdot\delta(\alpha,\beta)\cdot\delta(\delta,\epsilon)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta}\\ &+105\cdot R_z^4\cdot R_\gamma\cdot R_\zeta\cdot\delta(\alpha,\delta)\cdot\delta(\beta,\epsilon)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta} \end{aligned}$$

$$(1)$$

Simplify deltas:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta}=R_z^{-13}\cdot\frac{1}{315}\cdot$$

$$\begin{aligned} &+105\cdot R_z^4\cdot R_\beta\cdot R_\epsilon\cdot\delta(\alpha,\alpha)\cdot\delta(\gamma,\gamma)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\alpha\gamma\gamma\epsilon}\\ &+105\cdot R_z^4\cdot R_\beta\cdot R_\epsilon\cdot\delta(\alpha,\alpha)\cdot\delta(\gamma,\gamma)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\alpha\gamma\gamma\epsilon}\\ &+105\cdot R_z^4\cdot R_\beta\cdot R_\epsilon\cdot\delta(\alpha,\alpha)\cdot\delta(\gamma,\gamma)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\alpha\gamma\gamma\epsilon}\\ &+105\cdot R_z^4\cdot R_\beta\cdot R_\zeta\cdot\delta(\alpha,\alpha)\cdot\delta(\gamma,\gamma)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\alpha\gamma\gamma\zeta}\\ &+105\cdot R_z^4\cdot R_\beta\cdot R_\zeta\cdot\delta(\alpha,\alpha)\cdot\delta(\gamma,\gamma)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\alpha\gamma\gamma\zeta}\\ &+105\cdot R_z^4\cdot R_\beta\cdot R_\zeta\cdot\delta(\alpha,\alpha)\cdot\delta(\gamma,\gamma)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\alpha\gamma\gamma\zeta}\\ &+105\cdot R_z^4\cdot R_\gamma\cdot R_\delta\cdot\delta(\alpha,\alpha)\cdot\delta(\beta,\beta)\cdot\Theta_{\alpha\alpha}\cdot\Phi_{\beta\beta\gamma\delta}\\ &+105\cdot R_z^4\cdot R_\gamma\cdot R_\delta\cdot\delta(\alpha,\alpha)\cdot\delta(\beta,\beta)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\alpha\beta\gamma\delta}\\ &+105\cdot R_z^4\cdot R_\gamma\cdot R_\delta\cdot\delta(\alpha,\alpha)\cdot\delta(\beta,\beta)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\alpha\beta\gamma\delta}\\ &+105\cdot R_z^4\cdot R_\gamma\cdot R_\delta\cdot\delta(\alpha,\alpha)\cdot\delta(\beta,\beta)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\alpha\beta\gamma\delta}\\ &+105\cdot R_z^4\cdot R_\gamma\cdot R_\epsilon\cdot\delta(\alpha,\alpha)\cdot\delta(\beta,\beta)\cdot\Theta_{\alpha\alpha}\cdot\Phi_{\beta\beta\gamma\epsilon}\\ &+105\cdot R_z^4\cdot R_\gamma\cdot R_\epsilon\cdot\delta(\alpha,\alpha)\cdot\delta(\beta,\beta)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\alpha\beta\gamma\epsilon}\\ &+105\cdot R_z^4\cdot R_\gamma\cdot R_\epsilon\cdot\delta(\alpha,\alpha)\cdot\delta(\beta,\beta)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\alpha\beta\gamma\epsilon}\\ &+105\cdot R_z^4\cdot R_\gamma\cdot R_\epsilon\cdot\delta(\alpha,\alpha)\cdot\delta(\beta,\beta)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\alpha\beta\gamma\epsilon}\\ &+105\cdot R_z^4\cdot R_\gamma\cdot R_\zeta\cdot\delta(\alpha,\alpha)\cdot\delta(\beta,\beta)\cdot\Theta_{\alpha\alpha}\cdot\Phi_{\beta\beta\gamma\zeta}\\ &+105\cdot R_z^4\cdot R_\gamma\cdot R_\zeta\cdot\delta(\alpha,\alpha)\cdot\delta(\beta,\beta)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\alpha\beta\gamma\zeta} \end{aligned}$$

$$(2a)$$

$$\begin{aligned} & \xrightarrow{\text{cleaned up}} R_z^{-13} \cdot \frac{1}{315} \cdot \\ & +105 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\delta\epsilon} \\ & +105 \cdot R_z^4 \cdot R_\beta \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\gamma\delta} \\ & +105 \cdot R_z^4 \cdot R_\beta \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\gamma\delta} \\ & +105 \cdot R_z^4 \cdot R_\beta \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\gamma\delta} \\ & +105 \cdot R_z^4 \cdot R_\beta \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\gamma\epsilon} \\ & +105 \cdot R_z^4 \cdot R_\beta \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\gamma\epsilon} \\ & +105 \cdot R_z^4 \cdot R_\beta \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\gamma\epsilon} \\ & +105 \cdot R_z^4 \cdot R_\beta \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\gamma\zeta} \\ & +105 \cdot R_z^4 \cdot R_\beta \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\gamma\zeta} \\ & +105 \cdot R_z^4 \cdot R_\beta \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\gamma\zeta} \\ & +105 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\beta\beta\gamma\delta} \\ & +105 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\delta} \\ & +105 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\delta} \\ & +105 \cdot R_z^4 \cdot R_\gamma \cdot R_\epsilon \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\beta\beta\gamma\epsilon} \\ & +105 \cdot R_z^4 \cdot R_\gamma \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\epsilon} \\ & +105 \cdot R_z^4 \cdot R_\gamma \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\epsilon} \\ & +105 \cdot R_z^4 \cdot R_\gamma \cdot R_\zeta \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\beta\beta\gamma\zeta} \\ & +105 \cdot R_z^4 \cdot R_\gamma \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\zeta} \\ & +105 \cdot R_z^4 \cdot R_\gamma \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\zeta} \\ & +105 \cdot R_z^4 \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\beta\beta\delta\epsilon} \end{aligned} \quad (2b)$$

$$\begin{aligned}
& \xrightarrow{\text{added up}} R_z^{-13} \cdot \frac{1}{315} \cdot \left[\begin{aligned}
& +10395 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\delta\epsilon\zeta} \\
& -945 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\delta\epsilon\epsilon} \\
& -945 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\delta\delta\epsilon} \\
& -945 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\delta\delta\zeta} \\
& -945 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\gamma\delta\epsilon} \\
& -945 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\gamma\delta\zeta} \\
& -945 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\gamma\epsilon\zeta} \\
& -945 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\gamma\delta\epsilon} \\
& -945 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\gamma\delta\zeta} \\
& -945 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\gamma\epsilon\zeta} \\
& -945 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\delta\epsilon\zeta} \\
& -945 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\delta\epsilon} \\
& -945 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\delta\zeta} \\
& -945 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\epsilon\zeta} \\
& -945 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\delta\epsilon\zeta} \\
& -945 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\delta\epsilon\zeta} \\
& +315 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\gamma\gamma\delta\delta} \\
& +315 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\gamma\delta\delta} \\
& +315 \cdot R_z^4 \cdot R_\alpha \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\gamma\gamma\delta} \\
& +315 \cdot R_z^4 \cdot R_\alpha \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\gamma\gamma\epsilon} \\
& +315 \cdot R_z^4 \cdot R_\alpha \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\beta\gamma\gamma\zeta} \\
& +315 \cdot R_z^4 \cdot R_\beta \cdot R_\gamma \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\delta\delta} \\
& +315 \cdot R_z^4 \cdot R_\beta \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\gamma\delta} \\
& +315 \cdot R_z^4 \cdot R_\beta \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\gamma\epsilon} \\
& +315 \cdot R_z^4 \cdot R_\beta \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\gamma\zeta} \\
& +105 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\beta\beta\gamma\delta} \\
& +210 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\delta} \\
& +105 \cdot R_z^4 \cdot R_\gamma \cdot R_\epsilon \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\beta\beta\gamma\epsilon} \\
& +210 \cdot R_z^4 \cdot R_\gamma \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\epsilon} \\
& +105 \cdot R_z^4 \cdot R_\gamma \cdot R_\zeta \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\beta\beta\gamma\zeta} \\
& +210 \cdot R_z^4 \cdot R_\gamma \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\zeta} \\
& +105 \cdot R_z^4 \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\beta\beta\delta\epsilon} \\
& +210 \cdot R_z^4 \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\delta\epsilon} \\
& +105 \cdot R_z^4 \cdot R_\delta \cdot R_\zeta \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\beta\beta\delta\zeta} \\
& +210 \cdot R_z^4 \cdot R_\delta \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\delta\zeta} \\
& +105 \cdot R_z^4 \cdot R_\epsilon \cdot R_\zeta \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\beta\beta\epsilon\zeta} \\
& +210 \cdot R_z^4 \cdot R_\epsilon \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\epsilon\zeta} \\
& -45 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\beta\beta\gamma\gamma} - 180 \cdot R_z^6 \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\gamma}
\end{aligned} \right]
\end{aligned} \tag{2c}$$

Simplify R:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta}=R_z^{-13}\cdot\frac{1}{315}.$$

[illegible]

Multiply Out Sum:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta} = R_z^{-13} \cdot \frac{1}{315}.$$

[illegible]

$$\begin{aligned}
& \xrightarrow{\text{cleaned up}} R_z^{-13} \cdot \frac{1}{315} \cdot \left[\begin{aligned}
& +10395 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zzzz} - 5670 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{xxzz} \\
& - 5670 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{yyzz} - 5670 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zzzz} \\
& - 7560 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zzzz} - 945 \cdot R_z^6 \cdot \Theta_{xx} \cdot \Phi_{zzzz} \\
& - 945 \cdot R_z^6 \cdot \Theta_{yy} \cdot \Phi_{zzzz} - 945 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zzzz} \\
& + 315 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{xxxx} + 315 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{xxyy} \\
& + 315 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{xxzz} + 315 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{xyyy} \\
& + 315 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{yyyy} + 315 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{yyzz} \\
& + 315 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{xxzz} + 315 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{yyzz} \\
& + 315 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zzzz} + 1260 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{xxzz} \\
& + 1260 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{yyzz} + 1260 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zzzz} \\
& + 1260 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{xxzz} + 1260 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{yyzz} \\
& + 1260 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zzzz} + 630 \cdot R_z^6 \cdot \Theta_{xx} \cdot \Phi_{xxzz} \\
& + 630 \cdot R_z^6 \cdot \Theta_{xx} \cdot \Phi_{yyzz} + 630 \cdot R_z^6 \cdot \Theta_{xx} \cdot \Phi_{zzzz} \\
& + 630 \cdot R_z^6 \cdot \Theta_{yy} \cdot \Phi_{xxzz} + 630 \cdot R_z^6 \cdot \Theta_{yy} \cdot \Phi_{yyzz} \\
& + 630 \cdot R_z^6 \cdot \Theta_{yy} \cdot \Phi_{zzzz} + 630 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{xxzz} \\
& + 630 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{yyzz} + 630 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zzzz} \\
& + 1260 \cdot R_z^6 \cdot \Theta_{xx} \cdot \Phi_{xxzz} + 1260 \cdot R_z^6 \cdot \Theta_{yy} \cdot \Phi_{yyzz} \\
& + 1260 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zzzz} - 45 \cdot R_z^6 \cdot \Theta_{xx} \cdot \Phi_{xxxx} \\
& - 45 \cdot R_z^6 \cdot \Theta_{xx} \cdot \Phi_{xxyy} - 45 \cdot R_z^6 \cdot \Theta_{xx} \cdot \Phi_{xxzz} \\
& - 45 \cdot R_z^6 \cdot \Theta_{xx} \cdot \Phi_{xyyy} - 45 \cdot R_z^6 \cdot \Theta_{xx} \cdot \Phi_{yyyy} \\
& - 45 \cdot R_z^6 \cdot \Theta_{xx} \cdot \Phi_{yyzz} - 45 \cdot R_z^6 \cdot \Theta_{xx} \cdot \Phi_{xxzz} \\
& - 45 \cdot R_z^6 \cdot \Theta_{xx} \cdot \Phi_{yyzz} - 45 \cdot R_z^6 \cdot \Theta_{xx} \cdot \Phi_{zzzz} \\
& - 45 \cdot R_z^6 \cdot \Theta_{yy} \cdot \Phi_{xxxx} - 45 \cdot R_z^6 \cdot \Theta_{yy} \cdot \Phi_{xxyy} \\
& - 45 \cdot R_z^6 \cdot \Theta_{yy} \cdot \Phi_{xxzz} - 45 \cdot R_z^6 \cdot \Theta_{yy} \cdot \Phi_{xxyy} \\
& - 45 \cdot R_z^6 \cdot \Theta_{yy} \cdot \Phi_{yyyy} - 45 \cdot R_z^6 \cdot \Theta_{yy} \cdot \Phi_{yyzz} \\
& - 45 \cdot R_z^6 \cdot \Theta_{yy} \cdot \Phi_{yyzz} - 45 \cdot R_z^6 \cdot \Theta_{yy} \cdot \Phi_{yyzz} \\
& - 45 \cdot R_z^6 \cdot \Theta_{yy} \cdot \Phi_{zzzz} - 45 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{xxxx} \\
& - 45 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{xxyy} - 45 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{xxzz} \\
& - 45 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{xxyy} - 45 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{yyyy} \\
& - 45 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{yyzz} - 45 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{xxzz} \\
& - 45 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{yyzz} - 45 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zzzz} \\
& - 180 \cdot R_z^6 \cdot \Theta_{xx} \cdot \Phi_{xxxx} - 180 \cdot R_z^6 \cdot \Theta_{xx} \cdot \Phi_{xxyy} \\
& - 180 \cdot R_z^6 \cdot \Theta_{xx} \cdot \Phi_{xxzz} - 180 \cdot R_z^6 \cdot \Theta_{yy} \cdot \Phi_{xxyy} \\
& - 180 \cdot R_z^6 \cdot \Theta_{yy} \cdot \Phi_{yyyy} - 180 \cdot R_z^6 \cdot \Theta_{yy} \cdot \Phi_{yyzz} \\
& - 180 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{xxzz} - 180 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{yyzz} \\
& - 180 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zzzz}
\end{aligned} \right] \quad (5b)
\end{aligned}$$

$$\begin{aligned}
& \xrightarrow{\text{added up}} R_z^{-13} \cdot \frac{1}{315} \cdot \left[\begin{aligned}
& +720 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zzzz} - 2160 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{xxzz} \\
& - 2160 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{yyzz} - 360 \cdot R_z^6 \cdot \Theta_{xx} \cdot \Phi_{zzzz} \\
& - 360 \cdot R_z^6 \cdot \Theta_{yy} \cdot \Phi_{zzzz} + 270 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{xxxx} \\
& + 540 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{xxyy} + 270 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{yyyy} \\
& + 1620 \cdot R_z^6 \cdot \Theta_{xx} \cdot \Phi_{xxzz} + 540 \cdot R_z^6 \cdot \Theta_{xx} \cdot \Phi_{yyzz} \\
& + 540 \cdot R_z^6 \cdot \Theta_{yy} \cdot \Phi_{xxzz} + 1620 \cdot R_z^6 \cdot \Theta_{yy} \cdot \Phi_{yyzz} \\
& - 225 \cdot R_z^6 \cdot \Theta_{xx} \cdot \Phi_{xxxx} - 270 \cdot R_z^6 \cdot \Theta_{xx} \cdot \Phi_{xxyy} \\
& - 45 \cdot R_z^6 \cdot \Theta_{xx} \cdot \Phi_{yyyy} - 45 \cdot R_z^6 \cdot \Theta_{yy} \cdot \Phi_{xxxx} \\
& - 270 \cdot R_z^6 \cdot \Theta_{yy} \cdot \Phi_{xxyy} - 225 \cdot R_z^6 \cdot \Theta_{yy} \cdot \Phi_{yyyy}
\end{aligned} \right] \quad (5c)
\end{aligned}$$

Combining everything:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta} = \left[\begin{aligned} & +\frac{16}{7} \cdot R_z^{-7} \cdot \Theta_{zz} \cdot \Phi_{zzzz} - \frac{48}{7} \cdot R_z^{-7} \cdot \Theta_{zz} \cdot \Phi_{xxzz} \\ & - \frac{48}{7} \cdot R_z^{-7} \cdot \Theta_{zz} \cdot \Phi_{yyzz} - \frac{8}{7} \cdot R_z^{-7} \cdot \Theta_{xx} \cdot \Phi_{zzzz} \\ & - \frac{8}{7} \cdot R_z^{-7} \cdot \Theta_{yy} \cdot \Phi_{zzzz} + \frac{6}{7} \cdot R_z^{-7} \cdot \Theta_{zz} \cdot \Phi_{xxxx} \\ & + \frac{12}{7} \cdot R_z^{-7} \cdot \Theta_{zz} \cdot \Phi_{xxyy} + \frac{6}{7} \cdot R_z^{-7} \cdot \Theta_{zz} \cdot \Phi_{yyyy} \\ & + \frac{36}{7} \cdot R_z^{-7} \cdot \Theta_{xx} \cdot \Phi_{xxzz} + \frac{12}{7} \cdot R_z^{-7} \cdot \Theta_{xx} \cdot \Phi_{yyzz} \\ & + \frac{12}{7} \cdot R_z^{-7} \cdot \Theta_{yy} \cdot \Phi_{xxzz} + \frac{36}{7} \cdot R_z^{-7} \cdot \Theta_{yy} \cdot \Phi_{yyzz} \\ & - \frac{5}{7} \cdot R_z^{-7} \cdot \Theta_{xx} \cdot \Phi_{xxxx} - \frac{6}{7} \cdot R_z^{-7} \cdot \Theta_{xx} \cdot \Phi_{xxyy} \\ & - \frac{1}{7} \cdot R_z^{-7} \cdot \Theta_{xx} \cdot \Phi_{yyyy} - \frac{1}{7} \cdot R_z^{-7} \cdot \Theta_{yy} \cdot \Phi_{xxxx} \\ & - \frac{6}{7} \cdot R_z^{-7} \cdot \Theta_{yy} \cdot \Phi_{xxyy} - \frac{5}{7} \cdot R_z^{-7} \cdot \Theta_{yy} \cdot \Phi_{yyyy} \end{aligned} \right] \quad (6)$$